Speech, Language & Dysarthria Clinical Screening Summary

Report ID: SC-ASSESS-000001 | Assessment Date: October 27, 2026 at 03:30 UTC

Patient Identifier:	Sarah Jenkins	**Age & Biological Gender:**	48 yrs / Female
Neurological History:	None reported	**Symptom Onset / Duration**	Subacute (< 6 months)
Prior Assessments:	None recorded	**Speech Therapy History:**	None recorded
Evaluated Speech Task	Sustained Vowel	**Primary Concern:**	Speech screening evaluation

1. Severity Assessment of Dysarthria & Speech Production Analysis

DYSARTHRIA SEVERITY: TYPICAL / NONE

Intelligibility Index:	98.0% (Estimated)	**Speech Rate:**	240.0 WPM (6.8 syl/sec)
Articulation Precision Score:	98.5 / 100	**Acoustic Voice Quality:**	Normal Vocal Resonance
Pause Structure & Timing:	0 pauses (mean duration: 0.0	**Observed Characteristics:**	*Acoustic parameters within reference physiological ranges with intact speech rate, harmonic cla...*

2. Screening for Associated Speech & Language Disorders

Disorder Category	Screening Index (0-100%)	Clinical Diagnostic Status
Apraxia of Speech (AOS):	5.0% Index	Screening Indicator (Not a Confirmed Diagnosis)
Aphasia (Language Impairment):	5.0% Index	Screening Indicator (Requires Bedside Language Battery)
Stuttering / Disfluency:	5.0% Index	Screening Indicator (Disfluency Frequency Analysis)

3. Detailed Acoustic Biomarker Analysis & Signal Properties

Acoustic Biomarker	Measured Value	Reference Norms	Physiological Interpretation
Acoustic Speaker Gender	Female (70.6%)	Pitch F0 & Centroid	Vocal tract acoustic profile
Mean Pitch (F0)	175.5 Hz	M: 85-165Hz | F: 165-260Hz	Vocal fold vibrational frequency
Jitter (Local Perturbation)	0.20%	< 1.50% (Normal)	Pitch period vibration stability
Shimmer (Amplitude Perturbation)	0.50%	< 4.00% (Normal)	Glottal resistance & amplitude
Harmonics-to-Noise Ratio (HNR)	25.6 dB	> 15.0 dB (Normal)	Acoustic periodicity vs breathiness
Silence / Pause Dynamics	15.0%	10% - 25% (Typical speech)	Articulatory fluency & hesitations

4. Evidence-Based Speech Pathology Practice Suggestions

Targeted practice activities for clinical review by a licensed Speech-Language Pathologist.

- Resonant Voice Therapy (Forward Facial Mask Focus) (Vocal Wellness & Hygiene)

Instructions: Gently hum on an /m/ sound on an easy pitch. Place your fingers on your lips and nose to feel the buzz/vibration. Transition from 'Mmm' into 'Mmm-one', 'Mmm-two'.

Target:

- Pitch Siren Glides (Laryngeal Flexibility)

Instructions: Gently glide on an /oo/ sound from your comfortable speaking pitch upwards like a siren, then glide smoothly back down without straining.

Target:

5. Clinical Recommendations for Further Comprehensive Assessment

* Formal Motor Speech Evaluation by a licensed CCC-SLP (e.g., Frenchay Dysarthria Assessment-2 or Apraxia Battery for Adults-2).
* Comprehensive Language Assessment (e.g., Western Aphasia Battery-Revised or Boston Diagnostic Aphasia Examination) if word retrieval or syntax difficul
* Acoustic and Aerodynamic laryngeal videostroboscopy if dysphonia or vocal fold bowing is suspected.

SCREENING SUPPORT ONLY. REQUIRES FORMAL EVALUATION BY A LICENSED SPEECH-LANGUAGE PATHOLOGIST.

6. Clinician Review & Attestation Section

Review Status: PENDING FORMAL CLINICAL REVIEW **Reviewing Clinician: Not Assigned**

Clinician Notes: No clinician notes entered yet.

MANDATORY CLINICAL SCREENING & UNCERTAINTY DISCLAIMER:

SpeechCare AI is an assistive research and screening decision-support tool. It does NOT provide a definitive medical diagnosis. All screening indicators, severity classifications, and acoustic indices account for potential limitations in audio quality, microphone hardware, and environmental noise. These findings must NEVER replace a comprehensive bedside evaluation by a certified Speech-Language Pathologist (CCC-SLP) or licensed neurologist. Further clinical assessment is strongly advised before therapeutic or pharmacological interventions.